

Table: OTS generative VLMs are SOTA on image-to-text retrieval benchmarks.

Score	Method	ARO			
		Rel	Attr	COCO	Flickr
Random	-	50.0	50.0	20.0	20.0
Text-Only	Vera	61.7	82.6	59.8	63.5
	Grammar	59.6	58.4	74.3	76.3
$P_{LLM}(t)$	BART	81.1	73.6	95.0	95.2
	Flan-T5	84.4	76.5	98.0	98.2
	OPT	84.7	79.8	97.9	98.6
$P_{train}(t)$	BLIP	87.6	80.7	98.6	99.1
$P(match t, i)$	CLIP	59.0	62.0	46.0	60.0
	LAION2B-CLIP	51.6	61.9	25.2	30.2
	LAION5B-CLIP	46.1	57.8	26.1	31.0
	NegCLIP	81.0	71.0	86.0	91.0
	Structure-CLIP	83.5	85.1	-	-
	SyViC	80.8	72.4	92.4	87.2
	SGVL	-	-	87.2	91.0
	MosaiCLIP	82.6	78.0	87.9	86.3
	DAC-LLM	81.3	73.9	94.5	95.7
	DAC-SAM	77.2	70.5	91.2	93.9
	BLIP-ITC	63.1	81.6	34.3	41.7
	BLIP-ITM	58.7	90.3	45.1	51.3
$\frac{P_{train}(t i)}{P_{train}(t)^\alpha}$	Ours ($\alpha = 0$)	89.1	95.3	99.4	99.5
	Ours ($\alpha = 1$)	68.1	87.9	32.4	44.5
	Ours ($\alpha = \alpha^*$)	89.1	95.4	99.4	99.5

(a) Accuracy on ARO

Score	Method	SugarCrepe		
		Replace	Swap	Add
Random	-	50.0	50.0	50.0
Text-Only	Vera	49.5	49.3	49.5
	Grammar	50.0	50.0	50.0
$P_{LLM}(t)$	BART	48.4	51.9	61.2
	Flan-T5	51.4	57.6	40.9
	OPT	58.5	66.6	45.8
$P_{train}(t)$	BLIP	75.9	77.1	70.9
$P(match t, i)$	CLIP	80.8	63.3	75.1
	LAION2B-CLIP	86.5	68.6	88.4
	LAION5B-CLIP	85.0	68.0	89.6
	NegCLIP	88.3	76.2	90.2
	BLIP-ITC	85.8	73.8	85.7
	BLIP-ITM	88.7	81.3	87.6
$\frac{P_{train}(t i)}{P_{train}(t)^\alpha}$	Ours ($\alpha = 0$)	93.3	91.0	91.0
	Ours ($\alpha = 1$)	83.2	85.5	85.9
	Ours ($\alpha = \alpha^*$)	95.1	92.4	97.4

(c) Accuracy on SugarCrepe

Score	Method	VL-CheckList		
		Object	Attribute	Relation
Random	-	50.0	50.0	50.0
Text-Only	Vera	82.5	74.0	85.7
	Grammar	58.0	52.4	68.5
$P_{LLM}(t)$	BART	52.0	51.0	45.1
	Flan-T5	60.3	55.0	49.3
	OPT	59.3	48.8	60.0
$P_{train}(t)$	BLIP	68.2	58.7	75.9
$P(match t, i)$	CLIP	81.6	67.6	63.1
	LAION2B-CLIP	84.7	67.8	66.5
	LAION5B-CLIP	87.9	70.3	63.9
	NegCLIP	81.4	72.2	63.5
	SyViC	-	70.4	69.4
	SGVL	85.2	78.2	80.4
	SLVC	85.0	72.0	69.0
	DAC-LLM	87.3	77.3	86.4
	DAC-SAM	88.5	75.8	89.8
	BLIP-ITC	90.6	80.3	73.5
	BLIP-ITM	89.9	80.7	67.7
$\frac{P_{train}(t i)}{P_{train}(t)^\alpha}$	Ours ($\alpha = 0$)	92.6	78.7	90.8
	Ours ($\alpha = 1$)	90.4	77.6	77.8
	Ours ($\alpha = \alpha^*$)	94.4	82.1	92.8

(b) Accuracy on VL-CheckList

Score	Method	Crepe		
		Atom	Swap	Negate
Random	-	16.7	16.7	16.7
Text-Only	Vera	43.7	70.8	66.2
	Grammar	18.2	50.9	9.8
$P_{LLM}(t)$	BART	38.8	53.3	44.4
	Flan-T5	43.0	69.5	13.6
	OPT	53.3	72.7	5.0
$P_{train}(t)$	BLIP	55.4	69.7	60.8
$P(match t, i)$	CLIP	22.3	26.6	28.8
	LAION2B-CLIP	23.6	24.8	18.0
	LAION5B-CLIP	24.2	23.9	20.1
	BLIP-ITC	24.8	17.7	26.5
	BLIP-ITM	29.5	20.7	25.5
	Ours ($\alpha = 0$)	73.2	78.1	79.6
$\frac{P_{train}(t i)}{P_{train}(t)^\alpha}$	Ours ($\alpha = 1$)	20.6	28.3	35.6
	Ours ($\alpha = \alpha^*$)	73.3	78.1	79.6

(d) Accuracy on Crepe

LLM Generated Text

The proposed method ($\alpha = 0$ and $\alpha = \alpha^*$) consistently **outperforms all other baselines**, including text-only models, LLM-based probabilities, and CLIP-based matching, across all image-to-text retrieval benchmarks. Specifically, **it achieves state-of-the-art** accuracy on ARO, VL-CheckList, SugarCrepe, and Crepe.

Reference Text in the paper

We begin by evaluating blind language models **(in red)**. Surprisingly, this already produces SOTA accuracy on certain benchmarks such as AROFlickr, compared to the **best discriminative approaches (in gray)**. We also find that blind inference of generative VLMs, $P_{train}(t)$ via sampling Gaussian noise images **(in blue)**, **often performs better** and achieve **above-chance performance** even on the most recent Sugar Crepe. Next, we show that simply repurposing a generative VLM's language generation head for computing image-text scores (Visual GPT Score in yellow), which corresponds to $\alpha = 0$, **consistently produces** SOTA accuracy across all benchmarks. Finally, debiasing this score by tuning α on valset **(in green)** further **improves performance**, establishing the new SOTA.